

- 6 (a) there are no lost volts / energy lost in the battery
or there are no lost volts / energy lost in the internal resistance B1 [1]
- (b) the current / I decreases (as R increases) M1
p.d. decreases (as R increases) A1
- or
- the parallel resistance (of X and R) increases M1
p.d. across parallel resistors increases, so p.d. (across Y) decreases A1 [2]

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- (c) (i) current = 2.4 (A) C1
p.d. across AB = $24 - 2.4 \times 6 = 9.6 \text{ V}$ M1
- or
- total resistance = 10Ω (= $24 \text{ V} / 2.4 \text{ A}$) C1
(parallel resistance = 4Ω), p.d. = $24 \times (4 / 10) = 9.6 \text{ V}$ M1 [2]
- (ii) $R \text{ (AB)} = 9.6 / 2.4 = 4.0 \Omega$ C1
 $1/6 + 1/X = 1/4$ [must correctly substitute for R] C1
 $X = 12 \Omega$ A1
- or
- $I_R = 9.6 / 6.0 = 1.6 \text{ (A)}$ (C1)
 $I_X = 2.4 - 1.6 = 0.8 \text{ (A)}$ (C1)
 $X (= 9.6 / 0.8) = 12 \Omega$ (A1) [3]
- (iii) power = VI or EI or V^2/R or E^2/R or I^2R C1
= 24×2.4 or $(24)^2 / 10$ or $(2.4)^2 \times 10$
= 57.6 W (allow 2 or more s.f.) A1 [2]
- (d) power decreases M0
- e.m.f. constant or power = $24 \times$ current, and current decreases
or e.m.f. constant or power = $24^2 /$ resistance, and resistance increases A1 [1]